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THE INFLUENCE OF INFORMATION AND COMMUNICATION TECHNOLOGIES ON CONSUMER FREEDOM

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ABSTRACT

Objective: This study investigates how Information and Communication Technologies (ICTs) can restrict consumer freedom by influencing and changing their attitudes and behaviors. The aim is to understand how content personalization and algorithmic usage impact individual autonomy.

Theoretical Framework: The research draws on concepts from computer science, psychology, sociology, law, and ethics. It highlights key models and theories on content personalization, algorithmic manipulation, and unfair practices by major tech corporations, providing a comprehensive view of the impact of ICTs on consumer freedom.

Method: The methodological approach is deductive and multidisciplinary, involving an extensive literature review, case study analysis, and application of relevant theories. Methods such as document analysis and market practice investigation were used to assess the influence of ICTs on consumer autonomy.

Results and Discussion: The findings indicate that, although content personalization and algorithms are often viewed as beneficial, they can represent a subtle form of control and manipulation. The analysis revealed unfair practices, such as search result manipulation and data-driven consumer segmentation, which compromise consumer choice. These findings are discussed in light of the theoretical framework, addressing their implications and limitations.

Implications of the Research: Practical implications include the need for policies that protect consumers and promote ethical ICT usage. It is essential to regulate the use of personal data, ensure platform transparency, and foster critical digital literacy to mitigate the identified negative impacts.

Originality/Value: This study contributes to the literature by highlighting the relationship between ICTs and consumer autonomy in an innovative way. The research provides new insights into the subtle control exerted by technologies, underscoring the importance of regulations and more ethical practices in the field.

Keywords: Information and Communication Technologies, Consumer Control, Content Personalization, Digital Manipulation, Autonomy.

A INFLUÊNCIA DAS TECNOLOGIAS DA INFORMAÇÃO E COMUNICAÇÃO NA LIBERDADE DOS CONSUMIDORES

RESUMO

Objetivo: Este estudo investiga como as Tecnologias da Informação e Comunicação (TICs) podem restringir a liberdade dos consumidores, influenciando e alterando suas atitudes e comportamentos. O objetivo é entender como a personalização de conteúdo e o uso de algoritmos podem impactar a autonomia dos indivíduos.

Referencial Teórico: A pesquisa baseia-se em conceitos de ciência da computação, psicologia, sociologia, direito e ética. São destacados os principais modelos e teorias sobre personalização de conteúdo, manipulação de algoritmos e práticas desleais de grandes corporações tecnológicas, proporcionando uma visão abrangente do impacto das TICs na liberdade dos consumidores.

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Método: A abordagem metodológica é dedutiva e multidisciplinar, envolvendo uma revisão bibliográfica extensiva, análise de estudos de caso e aplicação de teorias relevantes. Foram utilizados métodos como análise documental e investigação de práticas de mercado para avaliar a influência das TICs na autonomia do consumidor.

Resultados e Discussão: Os resultados indicam que, embora a personalização de conteúdo e os algoritmos sejam frequentemente considerados benéficos, eles podem representar uma forma sutil de controle e manipulação. A análise revelou práticas desleais, como a manipulação de resultados de busca e segmentação baseada em dados pessoais, que comprometem a capacidade de escolha dos consumidores. Essas descobertas são discutidas à luz do referencial teórico, abordando suas implicações e limitações.

Implicações da Pesquisa: As implicações práticas incluem a necessidade de desenvolvimento de políticas que protejam os consumidores e promovam o uso ético das TICs. É fundamental regulamentar o uso de dados pessoais, garantir a transparência das plataformas e fomentar a alfabetização digital crítica para mitigar os impactos negativos identificados.

Originalidade/Valor: Este estudo contribui para a literatura ao destacar a relação entre TICs e autonomia do consumidor de maneira inovadora. A pesquisa oferece novas perspectivas sobre o controle sutil exercido pelas tecnologias, evidenciando a importância de regulamentações e práticas mais éticas no campo.

Palavras-chave: Tecnologias da Informação e Comunicação, Controle do Consumidor, Personalização de Conteúdo, Manipulação Digital, Autonomia.

LA INFLUENCIA DE LAS TECNOLOGÍAS DE LA INFORMACIÓN Y LA COMUNICACIÓN EN LA LIBERTAD DEL CONSUMIDOR

RESUMEN

Objetivo: Este estudio investiga cómo las Tecnologías de la Información y la Comunicación (TIC) pueden restringir la libertad del consumidor al influir y cambiar sus actitudes y comportamientos. El objetivo es comprender cómo la personalización de contenido y el uso algorítmico impactan la autonomía individual.

Marco teórico: La investigación se basa en conceptos de la informática, la psicología, la sociología, el derecho y la ética. Destaca los modelos y teorías clave sobre la personalización de contenido, la manipulación algorítmica y las prácticas desleales de las principales corporaciones tecnológicas, proporcionando una visión integral del impacto de las TIC en la libertad del consumidor.

Método: El enfoque metodológico es deductivo y multidisciplinario, e implica una extensa revisión de la literatura, análisis de estudios de caso y aplicación de teorías relevantes. Se utilizaron métodos como el análisis de documentos y la investigación de prácticas de mercado para evaluar la influencia de las TIC en la autonomía del consumidor.

Resultados y discusión: Los hallazgos indican que, aunque la personalización de contenido y los algoritmos a menudo se consideran beneficiosos, pueden representar una forma sutil de control y manipulación. El análisis reveló prácticas desleales, como la manipulación de los resultados de búsqueda y la segmentación de los consumidores basada en datos, que comprometen la elección del consumidor. Estos hallazgos se discuten a la luz del marco teórico, abordando sus implicaciones y limitaciones.

Implicaciones de la investigación: Las implicaciones prácticas incluyen la necesidad de políticas que protejan a los consumidores y promuevan el uso ético de las TIC. Es esencial regular el uso de datos personales, garantizar la transparencia de la plataforma y fomentar la alfabetización digital crítica para mitigar los impactos negativos identificados.

Originalidad/Valor: Este estudio contribuye a la literatura al destacar la relación entre las TIC y la autonomía del consumidor de una manera innovadora. La investigación proporciona nuevos conocimientos sobre el control sutil ejercido por las tecnologías, lo que subraya la importancia de las regulaciones y las prácticas más éticas en el campo.

Palabras clave: Tecnologías de la información y la comunicación, Control del consumidor, Personalización de contenido, Manipulación digital, Autonomía.

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1 INTRODUCTION

The Fourth Industrial Revolution, a concept widely discussed by Klaus Schwab (2016) in his work of the same name, represents a new era of technological innovation that is radically transforming society, the economy and the way people live and work. Characterized by the fusion of technologies that is blurring the lines between the physical, digital and biological spheres, this revolution is driven by advances in areas such as artificial intelligence, the internet of things (IoT), big data, robotics and biotechnology.

Information and Communication Technologies (ICTs) are a central component of this transformation. They not only facilitate global interconnection, but also enable the collection, analysis and use of data on an unprecedented scale. While these technologies offer countless opportunities for innovation and growth, they also raise critical questions about consumer freedom and autonomy. The ability of ICTs to influence and even manipulate behaviors and attitudes is an aspect that deserves in-depth analysis.

This article aims to explore how ICTs can restrict consumer freedom through inducing and changing attitudes and behaviors. The central premise is that, although these technologies offer a wide range of benefits, they also have a dark side that can compromise individuals' autonomy. This study seeks to examine the mechanisms through which ICTs exert this influence and discuss the ethical and social implications of this dynamic.

The justification for this study lies in the growing concern about privacy, data manipulation and the influence of large technological corporations on consumers. With the proliferation of connected devices and massive data collection, companies have the ability to create detailed consumer profiles and target content and advertising in a highly personalized way. This personalization, while often seen as beneficial, can lead to a subtle form of control and manipulation.

To achieve the proposed objectives, this study will adopt a multidisciplinary approach, combining *insights* from areas such as computer science, psychology, sociology, law and ethics. The methodology will include a comprehensive literature review, analysis of case studies and the application of relevant theories to understand the complex relationship between ICTs and consumer freedom.

The bibliographical review will identify and synthesize the main academic contributions on the topic, while the case studies will provide concrete examples of how ICTs influence consumer behavior. Furthermore, theories from behavioral psychology and ethics will be applied to analyze the implications of these practices and propose possible solutions to mitigate the negative effects.

The importance of this topic cannot be underestimated, given the central role that ICTs play in the Fourth Industrial Revolution. With the increasing digitalization of virtually every aspect of everyday life, from work to leisure, it is essential to understand how these technologies affect individuals' freedom and autonomy. Furthermore, as ICTs continue to evolve and become even more integrated into everyday life, the questions raised in this study will become increasingly relevant.

Understanding the mechanisms for inducing and changing behavior is essential to develop policies and practices that protect consumers and promote a more ethical and responsible use of ICTs. This study aims to contribute to this understanding, by offering a detailed and critical analysis of the ways in which ICTs can restrict consumers' freedom and proposing ways to mitigate these effects.

To fully understand the impact of ICTs on consumer freedom, it is useful to consider the historical context and evolution of these technologies within the Fourth Industrial Revolution. From the invention of the telegraph in the 19th century, through the development of the telephone, radio, television and, finally, the internet, each new wave of technological innovation has brought significant changes to the way people communicate and consume information.

The Fourth Industrial Revolution, however, goes beyond these previous innovations by integrating digital, physical and biological technologies in ways that were unimaginable just a few decades ago. The internet of things (IoT), for example, connects physical devices to the internet, allowing the collection and analysis of data in real time. Artificial intelligence and the *machine learning* allows systems to learn and adapt based on large volumes of data, creating new possibilities, but also new challenges in terms of controlling and manipulating behaviors.

ICTs present both challenges and opportunities. On the one hand, they offer powerful tools for education, communication and economic development. On the other hand, the ability to collect and analyze data on a large scale raises concerns about privacy, security and the potential manipulation of behavior.

One of the main challenges is finding a balance between beneficial personalization and harmful manipulation. Personalization can improve user experience by offering relevant content and products. However, when taken to the extreme, it can lead to a form of “filter

bubble", in which consumers are only exposed to information and products that reinforce their existing beliefs and preferences, in a way that limits their worldview and your ability to make informed decisions.

Ethical considerations are central to the discussion about the impact of ICTs on consumer freedom. Issues such as informed consent, transparency in data collection and use, and the responsibility of technology companies are fundamental to ensuring that ICTs are used in a way that respects the autonomy and dignity of individuals.

Furthermore, it is important to consider the social and economic implications of ICTs. The concentration of power in the hands of a few large technology companies can lead to inequalities and a form of control that compromises consumers' freedom. Therefore, it is essential to develop policies and regulations that promote fair competition and protect consumer rights.

This article aims to comprehensively and critically explore how ICTs can restrict consumers' freedom through inducing and changing behaviors. By analyzing the underlying mechanisms and ethical and social implications of this dynamic, we hope to contribute to a deeper understanding of the role of ICTs in contemporary society and propose paths for a more responsible and ethical use of these technologies.

The next section of the article will cover in detail how ICTs induce behaviors through algorithms and content personalization, providing a basis for subsequent discussion on changing consumer attitudes and behaviors.

2 ALGORITHMS AND CONTENT PERSONALIZATION

The informational revolution arises from the contrast between the machine tool revolution, which is based on the mechanization of manual functions, and the automation revolution, which is based on the mechanization of certain cognitive functions developed by industrial machinery. In this way, human work is redirected to the processing of information and communications (Pereira and Nascimento, 2016).

The rise of Information and Communication Technologies (ICTs) in the Fourth Industrial Revolution brought with it a series of innovations that transformed the way information and products are consumed. Among these innovations, algorithms and content personalization play a central role. These mechanisms, while often praised for improving the user experience, also raise significant concerns about consumers' freedom and autonomy.

Algorithms are sets of coded instructions that allow computer systems to perform specific tasks, such as recommending products or filtering information. In the context of ICTs, algorithms are widely used to personalize the content that users see on digital platforms, from social networks to streaming services and e-commerce sites. This personalization is based on the collection and analysis of large volumes of data about users' behaviors and preferences.

Eli Pariser (2012), in his work “The invisible filter: what the internet is hiding from you”, argues that personalization algorithms create a “filter bubble” by presenting users with only information and content that reinforces their beliefs and existing interests. This filter bubble can limit consumers' exposure to a diversity of information and perspectives, restricting their ability to make informed decisions and develop a broader worldview.

Content personalization not only limits the diversity of information available to consumers, but can also directly influence their consumption behaviors and decisions. Cathy O'Neil (2020) highlights that algorithms are often designed to maximize platforms' engagement and profit, often at the expense of users' autonomy. For example, recommendation algorithms on e-commerce platforms may direct consumers to products that generate the highest profit margin for the company, rather than those that best meet their needs.

Furthermore, algorithms can exploit consumers' psychological vulnerabilities. Content personalization can be used to create a sense of urgency or scarcity, encouraging impulsive purchases. This behavioral manipulation is particularly concerning when considering the amount of personal data that companies collect and analyze to optimize their marketing strategies.

Evgeny Morozov (2018) argues that large technology corporations, such as Google, Facebook and Amazon , have disproportionate power over consumer information and behavior. These companies use sophisticated algorithms to collect and analyze data at scale, allowing them to subtly but powerfully influence consumer decisions.

Morozov also highlights that this concentration of power in the hands of a few technology corporations raises significant ethical and political questions. The ability of these companies to shape consumer behavior and control the flow of information could have profound implications for democracy and individual autonomy. Content personalization, when used in a non-transparent manner and without users' informed consent, can be seen as a form of manipulation, which compromises consumers' freedom.

The ethical and social implications of content personalization are vast and complex. Manipulating behavior through algorithms raises questions about informed consent and transparency. Consumers are often unaware of how their data is collected and used, nor how

algorithms influence their decisions. This lack of transparency can undermine consumer trust in digital platforms and the companies that operate them.

Furthermore, content personalization can exacerbate existing inequalities. Cathy O'Neil (2020) argues that algorithms can perpetuate and amplify bias and discrimination, as they are often trained on historical data that reflects social inequalities. For example, credit algorithms may discriminate against ethnic minorities or disadvantaged socioeconomic groups, limiting their access to financial services.

To mitigate the negative effects of content personalization, it is essential to develop policies and practices that promote transparency, accountability and the protection of consumer rights. Some possible approaches include:

a) Transparency and Informed Consent: Companies must be transparent about how they collect and use consumer data and ensure that users provide informed consent for content personalization.

b) Regulation and Supervision: Governments and regulatory bodies must establish clear guidelines for the use of algorithms and content personalization, ensuring that companies' practices are fair and non-discriminatory.

c) Education and Awareness: It is important to educate consumers about how algorithms work and how their decisions can be influenced. This can empower users to make more informed decisions and protect their autonomy.

d) Development of Ethical Algorithms: Companies must invest in the development of algorithms that are fair, transparent and responsible, minimizing the potential for manipulation and discrimination.

Algorithms and content personalization play a central role in how ICTs influence consumer behavior in the Fourth Industrial Revolution. While they offer significant benefits, such as improved user experience, they also raise important concerns about consumers' freedom and autonomy. By addressing these issues in an ethical and responsible manner, it is possible to maximize the benefits of ICTs while minimizing their negative effects.

3 TARGETED ADVERTISING AND PREFERENCE MANIPULATION

From the beginning, the internet presented a challenging environment for traditional advertising. As already said, “the excess of advertisements forces specialists in the field to seek new forms of communication with the consumer, demanding increasingly radical movements to escape banality” (Rocha, 2009, p.83) . Audience saturation in traditional media has led

advertisers to explore the internet as a new route to access consumers. However, the internet's relevance to traditional advertising initially remained small.

One of the first advertising formats on the internet was the banner, “a message contained in a geometric figure that appears as soon as the internet user accesses a page” (Rocha, 2009, p.87) . Although innovative, the banner faced resistance from users, who quickly developed “banner blindness” by ignoring these visual ads.

The main characteristic that hindered the advertising use of the internet was the more active nature of internet users. It has already been observed that “the main characteristic of the internet that hinders its advertising use is the more active nature of the internet user in the face of messages and the medium itself” (Rocha, 2009, p.88) . Unlike traditional media, in which the consumer is a passive receiver, on the internet, the user has control over the content they consume and can easily avoid unwanted advertisements.

This Internet user selectivity required advertisers to develop more sophisticated and less invasive strategies. Advertising had to evolve into forms that offered value to the consumer, such as sponsored content, native ads and influencer marketing.

Over time, internet advertising has undergone several transformations. The introduction of targeting and personalization technologies has allowed ads to be targeted at specific audiences based on their interests and *online behaviors* . This increased the relevance of ads and improved the user experience.

Furthermore, the emergence of social networks has revolutionized *online advertising* . Platforms like Facebook, Instagram and Twitter offer robust tools for audience segmentation and performance measurement, enabling more effective and measurable campaigns.

Another significant innovation was the advent of content *marketing and* influencer *marketing* . These strategies focus on creating valuable, relevant content that attracts and engages audiences, rather than interrupting them with intrusive ads. Digital influencers, with their loyal follower bases, have become powerful channels for promoting products and services.

The internet offers advertisers a dynamic and interactive environment for consumer research. Cathy O'Neil (2020) highlights that *feedback* from each advertising release arrives in seconds, allowing campaigns to be adjusted in hours, instead of months, as occurred with traditional methods such as mail. This speed and efficiency are possible thanks to the collection and analysis of large volumes of data, which make it possible to identify consumer behavior patterns and preferences.

One of advertising's brightest promises is reaching the right consumer, at the right time, with the most effective message. ICTs allow this promise to become increasingly closer to

reality. The personalization of advertising campaigns is one of the main benefits provided by ICTs. Using data collected from a variety of sources, such as browsing history, social media interactions and geographic location, advertisers can target consumers into specific groups and target highly relevant messages.

The author notes that this fine-tuning of advertising campaigns never stops, as machines are constantly examining our data, looking for habits, hopes, fears and desires (O"Neil , 2020). With the use of machine learning , a growing domain of artificial intelligence, computers can dive into data by following just basic instructions, finding patterns on their own and tracing relationships between those patterns and results over time.

The persuasive power of ICTs in advertising lies in the ability to personalize and adjust messages continuously and precisely. By identifying patterns of behavior and preferences, algorithms can predict which messages are most likely to provoke a purchasing decision. This persuasiveness is amplified by the speed at which data is processed and analyzed, allowing advertisers to quickly respond to changing consumer preferences.

Furthermore, the personalization of advertising messages can exploit consumers' emotional and psychological vulnerabilities. By understanding individuals' fears and desires, advertisers can create messages that resonate deeply with their emotions to encourage impulsive or unnecessary purchases. This subtle but powerful manipulation raises ethical questions about consumer autonomy and the transparency of advertising practices.

The ethical and social implications of using ICTs in advertising are vast and complex. Manipulating preferences through algorithms raises questions about informed consent and transparency. Consumers are often unaware of how their data is collected and used, nor how algorithms influence their decisions. This lack of transparency can undermine consumer trust in digital platforms and the companies that operate them.

Furthermore, the personalization of advertising campaigns can perpetuate and amplify prejudices and inequalities. Cathy O'Neil (2020) argues that advertising algorithms can discriminate against certain demographic groups by displaying different ads based on characteristics such as race, gender, or socioeconomic *status* . This can lead to a form of digital segregation, where different groups of consumers have access to different products and services, exacerbating existing inequalities.

4 UNFAIR PRACTICES OF BIG DATA

Given the great power concentrated in the hands of a few companies, consumers – and even competitors – are subject to abusive practices by these technology companies. Google has been accused of manipulating search results to favor its own services over competitors. The French Competition Authority denounced this practice, in order to highlight the possible manipulation of results indices (Behar- Touchais , 2020) .

Imagine that a consumer performs a search on a search engine. The probability of consumers looking for companies that appear in the first results is greater than those that appear, for example, in 123rd place.

The company Kelkoo was accused of not clearly identifying its role as an advertiser on its price comparison website, which could mislead consumers.

Article L. 442-6, item I, paragraph 5 of the French Commercial Code allows a de-indexed advertiser to be compensated for the period of reasonable notice. However, this measure is considered inadequate to deal with abusive deindexation (Behar- Touchais , 2020) . The company E- Kanopi , which operated several websites and had AdWords and AdSense accounts with Google, was deindexed without a clear justification, exemplifying a case of abuse of market power by Google. Google was accused of abusing its dominant position in the market, harming the competitive functioning of the market and favoring its own services (Behar-Touchais , 2020) .

As you can see, all these unfair competition practices, by reducing the consumer's ability to choose, also harm them.

5 SOCIAL NETWORKS AND THE INFLUENCE ON CONSUMER DECISIONS

Social networks have become powerful influencers on consumer behavior, shaping consumer decisions and relationships with brands. Precise audience segmentation through data collection and sophisticated algorithms enable the personalization of targeted ads and content to directly impact purchasing choices (Efendioğlu , 2019). However, this algorithmic segmentation can create a “filter bubble” to limit exposure to different perspectives and to perpetuate biases, leading to standardized consumption (Can and Engindeniz , 2012).

The presence of digital influencers on social networks amplifies the power of brands, in order to organically integrate products and services into the content consumed by users, often leading to impulsive and less than rational purchasing decisions. Furthermore, the search for

social validation on social networks, through likes and shares, drives users to consume as a way of gaining recognition and belonging, which feeds a cycle of ostentatious consumption (Can and Engindeniz , 2012).

Recommendations and opinions from friends and family on social media can have a significant impact on consumers' purchasing decisions. Interactions with brands on social media, such as likes, comments and shares, can influence consumers' attitudes towards the brand (Can and Engindeniz , 2012). Consumers who contribute brand-related content, such as reviews and posts, can influence other consumers and create a community around the brand. User-generated content is often seen as more authentic and trustworthy than traditional advertising, which can increase consumers' trust in the brand. (Can and Engindeniz , 2012).

consumers' *offline* purchasing decisions . To guarantee consumer autonomy amid this hyperstimulation and algorithmic manipulation, it is essential to promote awareness about the influence mechanisms present on social networks (Gelinas , 2017). Furthermore, regulating the collection and use of data by digital platforms is essential to protect users' privacy and autonomy.

Rethinking the business model of social networks, in order to prioritize the well-being and autonomy of users, is an urgent challenge to be faced to build a more ethical and transparent digital environment (Mertz , 2023). Ultimately, understanding and questioning the mechanisms of influence on social media is essential to guarantee consumer autonomy and promote a fairer and more democratic digital future.

6 THE AUTONOMY OF WILL AT RISK IN PLATFORM CAPITALISM

The advent of platform capitalism ushered in a new era in consumption, marked by the ubiquity of data and the promise of personalized experiences. Digital platforms such as Amazon , Google and Facebook connect users and suppliers, shaping a complex ecosystem that, under the facade of convenience and efficiency, raises questions about consumer autonomy. Based on the massive collection of data and the application of persuasion techniques, these platforms exercise a subtle but significant power in consumer decision-making, challenging the classic notion of autonomy of will and encouraging impulsive consumption and, at times, irrational.

The architecture of choice, a concept explored by Thaler and Sunstein (2018, p.136) in their book " Nudge ", illustrates how the organization of options presented influences decisions, even if unconsciously.

> architecture system is one that helps people improve their ability to map and, consequently, select options that can improve their well-being. One of the ways to do this is to make the information about each of the options more understandable and transform the numerical data into units that can be applied more easily and quickly.

Digital platforms, using browsing data, purchase history and social interactions, personalize the user experience, directing them to pre-determined choices. Sophisticated algorithms predict preferences and behaviors, creating “filter bubbles” that limit exposure to products and diverse information, undermining consumers’ critical capacity and freedom of choice (Pariser , 2012).

Manipulating the perception of urgency and scarcity is another strategy used to encourage impulsive consumption. Messages such as “last unit” or “limited time offer” exploit the fear of loss and risk aversion, leading to hasty and less than rational decisions (Cialdini , 1990). Additionally, the ubiquity of online advertising, often personalized and integrated into content in an imperceptible way (*native advertising*), contributes to informational saturation and the difficulty in discerning between informative and persuasive content.

Gamification, in turn, uses game elements to increase consumer engagement and loyalty. Points, rankings and rewards, although apparently harmless, exploit the brain's reward system, leading to the search for instant gratification and the formation of compulsive consumption habits (Kim and Werbach , 2016). The feeling of pleasure associated with obtaining virtual rewards masks the real costs of consumption, encouraging a vicious cycle of unnecessary purchases.

Furthermore, the attention economy, a hallmark of platform capitalism, intensifies the competition for the user's time and attention. Constant notifications, endless news feeds and the promise of instant gratification contribute to dispersion, anxiety and reduced ability to concentrate (NEWPORT, 2016). Fragmented attention makes the consumer more susceptible to emotional appeals and less likely to make conscious and considered choices.

The ethical implications of manipulating choice in the digital environment are profound. By undermining consumer autonomy, digital platforms erode the basis of individual responsibility, a fundamental element for conscious and sustainable consumption. Furthermore, extreme personalization and the creation of “filter bubbles” can contribute to social polarization and the spread of misinformation, with negative impacts on democracy and the social fabric.

It is essential to raise awareness about the persuasion mechanisms used by digital platforms and promote critical digital literacy. Consumer education on the responsible use of digital technologies, regulation of the use of personal data and the promotion of transparency

by platforms are essential measures to guarantee autonomy of will and the construction of a fairer and more balanced digital ecosystem. After all, the promise of convenience and personalization cannot override the fundamental right to make free and conscious choices.

7 CONCLUSION

The Fourth Industrial Revolution, driven by Information and Communication Technologies (ICTs), has radically transformed society, the economy and everyday life. While these technologies offer countless opportunities for innovation and growth, they also raise critical questions about consumer freedom and autonomy. This study explored how ICTs can restrict consumers' freedom through inducing and changing attitudes and behaviors, highlighting the underlying mechanisms and ethical and social implications of these dynamics.

Extreme personalization and the creation of “filter bubbles” are two of the main mechanisms through which ICTs can influence consumers. While personalization can be seen as beneficial, it can also lead to a subtle form of control and manipulation, limiting individuals' worldview and compromising their ability to make informed decisions. Furthermore, the massive collection of data and the ability of companies to create detailed profiles of consumers raise significant concerns about privacy and data manipulation.

To mitigate these negative effects, it is essential to develop policies and practices that protect consumers and promote a more ethical and responsible use of ICTs. Raising awareness about the persuasion mechanisms used by digital platforms and promoting critical digital literacy are fundamental steps. Consumer education on the responsible use of digital technologies, regulation of the use of personal data and the promotion of transparency by platforms are essential measures to guarantee autonomy of will and the construction of a fairer and more balanced digital ecosystem.

Ultimately, the promise of convenience and personalization of ICTs cannot override the fundamental right of individuals to make free and informed choices. This study hopes to contribute to a deeper understanding of the role of ICTs in contemporary society and propose paths for a more responsible and ethical use of these technologies. By addressing the issues raised, we can work towards a future in which technological innovation and individual freedom coexist harmoniously.

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